

* Oefeningen reeks 8:

1.

	x	y	z
S	xSx	ySy	zSz

2. Twee, want als we een lookahead symbool gebruiken dan weten we niet welke productie regel we moeten gebruiken.

3.

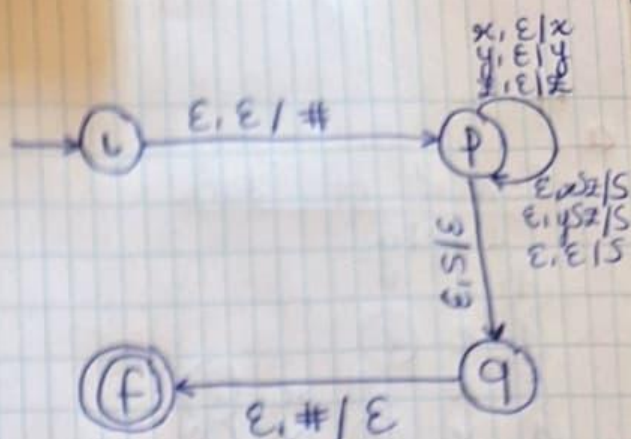
$$P = (Q, \Sigma, \Gamma, \delta, \iota, \epsilon, \{f\})$$

$$Q = \{\iota, f, p, q\}$$

$$\Sigma = \{x, y, z\}$$

$$\Gamma = \{x, y, z, S, \#\}$$

Transitie Functie δ



	ε, ε	ε, #	x, ε	y, ε	z, ε	ε, xSx	ε, ySy	ε, zSz	ε, S
→ l	p, #								
q		f, ε							
p			p, x	p, y	p, z	p, S	p, S	p, S	q, ε
* f									

4.

a.

Content of stack

empty

①

① x ②

① x ② x ③

① x ② x ③

① x ③

① S ⑨

accept!

Remaining input

xx

xx

x

2.

b.

empty

①

① x ②

① x ② x ③

① x ② x ③ y ④

① x ② Y ④

① x ② Y ④ y ⑤

① Y ⑦

① S ⑨

accept!

xx yy
xx yy
x yy
yy
yy
y

* oefeningen reeks 9

1.

a.

$q_0 \cdot 00 \vdash Xq_1 0 \vdash X0q_1$

b.

$q_0 000 111 \vdash Xq_1 00 111 \vdash X0q_1 0 111$

$\vdash X0q_1 111 \vdash X^1 0q_1 0Y11 \vdash Xq_1^1 00Y11$

$\vdash q_1 X00Y11 \vdash Xq_1^2 00Y11 \vdash XXq_1^2 0Y11$

$\vdash XX0q_1^2 Y11 \vdash XX0^2 Yq_1^2 11 \vdash XX0q_1^2 YY1$

$\vdash XXq_1^2 0Y11 \vdash Xq_1^2 X0YY1$

$\vdash XXq_1^2 0Y11 \vdash XXXq_1^2 YY1 \vdash XXXYq_1^2 Y1$

$\vdash XXX^2 YYq_1^2 1 \vdash XXX^2 Yq_1^2 YY \vdash XXXq_1^2 YY$

$\vdash XXq_1^2 xYY1 \vdash XXXq_1^2 YY1 \vdash XXXYq_1^2 YY$

$\vdash XXXYYq_1^2 Y \vdash XXXYYXq_1^2 B \vdash XXXYYBYq_1^4$

2. $M = (\{q_0, q_1, q_2, q_3, q_4\}, \{0, 1\}, \{0, 1, B\}, \delta, q_0, B, \{q_4\})$

$\delta =$	0	1	B
$\rightarrow q_0$	(q_1, B, R)	(q_3, B, R)	(q_4, B, R)
q_1	$(q_1, 0, R)$	(q_2, B, L)	(q_1, B, R)
q_2	$(q_2, 0, L)$	$(q_2, 1, L)$	(q_0, B, R)
q_3	(q_3, B, L)	$(q_3, 1, R)$	(q_3, B, R)
q_4	(q_4, B, R)	(q_3, B, R)	